

Sales Exploratory Data Analysis (EDA) Report

This report presents an exploratory data analysis (EDA) conducted on a sales dataset containing transactional information related to products, referral sources, payment methods, coupon usage, customer purchases, and revenue performance.

1. Project Objective

The objective of this project was to analyze the sales dataset to identify patterns, trends, distributions, and outliers that influence overall business performance and customer behavior.

2. Dataset Description

The dataset contains transactional sales information including products, revenue, referral sources, payment methods, coupon usage, customer IDs, quantities, and order statuses.

3. Descriptive Statistics

Metric	Value
Total Orders	1,200
Average Order Value	\$1,053.97
Median Order Value	\$823.62
Maximum Order Value	\$3,456.40
Minimum Order Value	\$11.39
Total Revenue	\$1,264,761.96

4. Key Analysis Performed

- Revenue trend analysis
- Product performance analysis
- Referral source analysis
- Payment method distribution analysis
- Order status distribution analysis
- Outlier detection for high-value transactions

5. Key Findings

- Instagram generated the highest referral revenue at \$275,285.45.
- Credit Card transactions generated the highest payment method revenue at \$263,847.63.

- Chair and Printer products generated the highest product revenues exceeding \$195,000 each.
- Several high-value transactions above \$3,400 were identified as revenue outliers.
- The dataset generated a total revenue of \$1,264,761.96 across 1,200 orders.
- Cancelled and returned orders represented a significant share of operational activity.

6. Revenue by Product

Product	Revenue
Chair	\$195,620.11
Printer	\$195,612.61
Laptop	\$192,126.56
Tablet	\$186,568.95
Monitor	\$175,651.41
Desk	\$167,459.93
Phone	\$151,722.39

7. Outlier Analysis

Several unusually large transactions were identified during the analysis. The maximum order value of \$3,456.40 significantly exceeded the median order value of \$823.62, indicating the presence of premium customer purchases and bulk-order behavior.

8. Business Insights

The analysis indicates that Instagram and digital payment methods contributed strongly to revenue generation. Product sales were relatively diversified across categories, while several high-value transactions contributed disproportionately to overall revenue.

9. Tools Used

- Microsoft Excel
- PivotTables and PivotCharts
- Descriptive Statistics

10. Conclusion

The exploratory data analysis successfully identified important trends, patterns, and operational insights within the sales dataset. The findings provide valuable information for improving marketing strategies, customer engagement, and business decision-making.